SAT Math Mastery · Session 1

Quadratics and Nonlinear Functions

Time	3 hours: Part A in class (140 min solving) + 40 min self-check and error log. Part B is the self-assessment, done independently.
Topics	parabolas: vertex, zeros and intercepts · function notation and composition · transformations · exponential and power functions · direct, inverse and joint variation · equivalent forms · undefined values
Name / Date	_____ _____

PART A · IN CLASS — 35 PROBLEMS

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1. HARD · DESMOS

The function g satisfies $g(kn) = n + 10$ for every real number n, where k is a constant. If $g(4) = 82$, what is the value of k?

2. HARD · DESMOS

A decorative fountain in a public park shoots a jet of water whose height above the ground, in meters, is modeled by

$$h(t) = -4\,(t-3)^2 + 45,$$

where t is the time, in seconds, after the jet is activated. For which values of t does the water reach a height of 29 meters?

(A) $t = 5$ only

(B) $t = 1$ and $t = 5$

(C) $t = 2$ and $t = 4$

(D) $t = 1$ only

3. HARD · DESMOS

The function f is defined by

$$f(x) = a\,(x-2)\,(x+6)\,,$$

where a is a constant. The graph of $y = f(x)$ in the xy-plane passes through the point $(0, -24)$. What is the value of $f(5)$?

4. HARD · DESMOS

The function f is defined by $f(x) = a \cdot 2^x$, where a is a constant. If $f(3) = 40$, what is the value of $f(5)$?

(A) 64

(B) 128

(C) 160

(D) 200

5. HARD · DESMOS

A ball is thrown upward, and its height above the ground, in meters, is modeled by

$$h(t) = -5\,(t-2)^2 + 30,$$

where t is the time, in seconds, after the ball is thrown. For which values of t does the ball reach a height of 10 meters?

(A) $t = 0$ and $t = 4$

(B) $t = 1$ and $t = 3$

(C) $t = 0$ only

(D) $t = 4$ only

6. HARD · DESMOS

The function h is defined by $h(x) = (x - a)(x - b)$, where a and b are constants and $a < b$. The graph of $y = h(x)$ crosses the x-axis at $x = -3$ and $x = 7$. What is the minimum value of h?

(A) -20

(B) -10

(C) 25

(D) -25

7. HARD · DESMOS

The function p is defined by $p(x) = a \cdot x^3$, where a is a nonzero constant. If $p(2) = 40$, what is the value of $p(-2)$?

8. HARD · DESMOS

The function N is defined by $N(t) = k \cdot 2^{-at}$ for positive constants k and a. The table shows values of N for select values of t.

t	0	1	2
$N(t)$	200	50	12.5

What is the value of a?

9. Hard · Desmos

The variable z varies jointly with x and the square of y. If $z = 48$ when $x = 3$ and $y = 4$, what is the value of z when $x = 5$ and $y = 2$?

(A) 10

(B) 16

(C) 20

(D) 25

10. Hard · Desmos

The graph of $y = h(x)$ is the graph of $y = x^2$ shifted right 3 units and down 5 units. What is the value of $h(7)$?

(A) 6

(B) 11

(C) 14

(D) 16

11. Hard · Desmos

A ball's height above the ground, in meters, t seconds after being launched from a platform is modeled by $h(t) = -5t^2 + 20t + c$, where c is the platform height. If $h(1) = 35$, what is the value of $h(3)$?

12. HARD · DESMOS

The function g is defined by

$$g(x) = a\,(x-3)\,(x+4)\,,$$

where a is a constant. The graph of $y = g(x)$ in the xy-plane passes through the point $(0, -36)$. What is the value of $g(6)$?

13. HARD · DESMOS

A cooperative tracks its annual export tonnage with the function $f(x) = a \cdot 3^x$, where a is a constant and x is the number of years since record-keeping began. If $f(2) = 54$, what is the value of $f(4)$?

(A) 108

(B) 162

(C) 324

(D) 486

14. HARD · DESMOS

The height, in meters, of a ball above the ground t seconds after being thrown is given by $h(t) = -4.9t^2 + 14.7t + 2$. Which of the following is closest to the number of seconds it takes for the ball to hit the ground ($h(t) = 0$)?

(A) 3.1

(B) 3.6

(C) 2.1

(D) 2.6

15. Hard · Desmos

The graph of the function $g(x) = (x - 3)^2 - 7$ is the same as the graph of $y = x^2$ after which of the following transformations?

(A) Shifted left 3 and up 7

(B) Shifted right 3 and down 7

(C) Shifted right 3 and up 7

(D) Shifted left 3 and down 7

16. Hard · Desmos

If $x = 5$ is one solution to the equation $x^2 - kx + 20 = 0$, what is the value of k?

17. Hard · Desmos

A company's weekly revenue, in thousands of dollars, from selling a product priced at x dollars is modeled by $R(x) = -3x^2 + 150x$. What price x maximizes the weekly revenue?

(A) 15

(B) 20

(C) 25

(D) 30

18. HARD · DESMOS

The equation $x^2 - 12x + c = 0$ has two equal (repeated) roots. What is the value of c?

19. HARD · DESMOS

The function f is defined by $f(x) = \dfrac{12 - x}{2 + x}$, where $x \neq -2$. As the value of x increases without bound, the value of $f(x)$ approaches which of the following?

(A) -1

(B) 1

(C) 12

(D) -2

20. HARD · DESMOS

$y = (x - p)^2 (x + p)(x - q)$, where p and q are positive constants and $p \neq q$. How many distinct x-intercepts does the graph of this equation have in the xy-plane?

(A) 4

(B) 1

(C) 2

(D) 3

21. HARD · DESMOS

The table below shows some values of the function N.

t	0	1	2
$N(t)$	100	25	$\frac{25}{4}$

If $N(t) = k \cdot 2^{-at}$ for positive constants k and a, what is the value of a?

22. HARD · DESMOS

The function f is defined by $f(x) = 3x + k$, where k is a constant. If $f(f(2)) = 38$, what is the value of k?

23. HARD · DESMOS

The function f is defined by $f(x) = a(x+4)(x-10)$, where a is a constant. The minimum value of f is -98. What is the value of $f(0)$?

(A) 80

(B) -98

(C) -80

(D) -40

24. Hard · Desmos

A theater's revenue, in dollars, from selling t tickets is $R(t) = t(320 - 10t)$. For which of the following values of t is the revenue 2,400 dollars?

(A) 18

(B) 20

(C) 26

(D) 14

25. Hard · Desmos

$y = \dfrac{5(x+3)(x-9)(x+2)}{(x-5)(x-5c)}$ In the equation above, c is a positive constant. The expression is undefined at $x = 10$ (and $x = 5$). What is the value of c?

(A) 3

(B) -3

(C) -2

(D) 2

26. Hard · Desmos

The function $f(x) = ax^2 + bx + c$ has $a > 0$ and $c < 0$. Which of the following must be true?
I. f has two distinct real zeros
II. The zeros of f have opposite signs
III. The vertex of the graph of f lies in the fourth quadrant

(A) I and II only

(B) I, II and III

(C) I only

(D) I and III only

27. HARD · DESMOS

x	$g(x)$
-12	$-\frac{17}{4}$
-9	$-\frac{10}{3}$
3	258

The table above shows three values of x and their corresponding values of $g(x)$. If $g(x) = \frac{x^2 + a}{bx} + 2^{(cx-1)}$, what is the value of abc?

28. HARD · DESMOS

The function f is defined by $f(x) = 5(3)^{x+2}$. The functions g and h are equivalent to f. Which of the following equations displays, as a constant or coefficient, the y-coordinate of the y-intercept of the graph of the function it defines in the xy-plane?
I. $g(x) = 45(3)^x$
II. $h(x) = 15(3)^{x+1}$

(A) Neither I nor II

(B) I only

(C) II only

(D) I and II

29. Hard · Desmos

The function f is defined by $f(x) = 96(2)^{x-3}$. The functions g and h are equivalent to f. Which of the following equations displays, as a constant or coefficient, the y-coordinate of the y-intercept of the graph of the function it defines in the xy-plane?
I. $g(x) = 48(2)^{x-2}$
II. $h(x) = 12(2)^{x}$

(A) Neither I nor II

(B) I only

(C) II only

(D) I and II

30. Hard · Desmos

The function f is defined by $f(x) = 3(4)^{x+2}$, where $x \geq 0$. The functions g and h are equivalent to f. Which of the following equations displays, as a constant or coefficient, the minimum value of the function it defines, where $x \geq 0$?
I. $g(x) = 48(4)^{x}$
II. $h(x) = 48(2)^{2x}$

(A) I and II

(B) Neither I nor II

(C) I only

(D) II only

31. Hard · Desmos

The function f is defined by $f(x) = 6(2)^{x+3}$, where $x \geq 0$. The functions g and h are equivalent to f. Which of the following equations displays, as a constant or coefficient, the minimum value of the function it defines, where $x \geq 0$?
I. $g(x) = 12(2)^{x+2}$
II. $h(x) = 24(2)^{x+1}$

(A) I only

(B) II only

(C) I and II

(D) Neither I nor II

32. Hard · Desmos

The function $f(x) = 200(1.5)^{\frac{x}{2}}$ is defined for all x. For every increase of x by 4, the value of $f(x)$ increases by $p\%$. What is the value of p?

33. Hard · Desmos

The function $f(t) = 2400(0.8)^{\frac{t}{5}}$ models the value, in dollars, of a machine t years after it was bought. For every increase of t by 10, the value of $f(t)$ decreases by $p\%$. What is the value of p?

(A) 64

(B) 20

(C) 36

(D) 40

34. HARD · DESMOS

The function $N(t) = 500(2)^{\frac{t}{3}}$ models the number of bacteria in a culture t hours after the start. Which of the following gives the number of bacteria m minutes after the start?

(A) $N(m) = 500(2)^{\frac{m}{20}}$

(B) $N(m) = 500(2)^{\frac{m}{180}}$

(C) $N(m) = 500(2)^{180m}$

(D) $N(m) = 500(2)^{20m}$

35. HARD · DESMOS

The function $f(x) = 12{,}000(0.85)^x$ models the number of visitors a museum had each year, where x is the number of years since the start of 2018 and $0 \leq x \leq 6$. If $y = f(x)$ is graphed in the xy-plane, which of the following is the best interpretation of the y-intercept of the graph in this context?

(A) The minimum estimated number of visitors during the 6 years was 4,526.

(B) The minimum estimated number of visitors during the 6 years was 12,000.

(C) The estimated number of visitors in 2018 was 4,526.

(D) The estimated number of visitors in 2018 was 12,000.

PART B · SELF-ASSESSMENT — 23 PROBLEMS

36. Medium · Desmos

The function h is defined by $h(x) = 2x^2 - 12x + 7$. What is the x-coordinate of the vertex of the graph of $y = h(x)$ in the xy-plane?

(A) 3

(B) 6

(C) -3

(D) 2

37. Medium · Desmos

The function f is defined by $f(x) = a \cdot 3^x$, where a is a constant. If $f(2) = 45$, what is the value of $f(4)$?

(A) 225

(B) 405

(C) 450

(D) 135

38. Medium · Desmos

If $f(x) = x^2 - 4x + k$ and $f(1) = 9$, what is the value of k?

(A) 6

(B) 9

(C) 12

(D) 14

39. Medium · Desmos

Let $f(x) = ax^2$ and $g(x) = bx^4$ for any value of x, where a and b are positive constants. For how many values of x is $f(x) = g(x)$?

(A) Three

(B) None

(C) One

(D) Two

40. Medium · Desmos

The variable y varies inversely with x. When $x = 6$, $y = 15$. What is the value of y when $x = 10$?

(A) 25

(B) 5

(C) 6

(D) 9

41. Medium · Desmos

If $f(x) = x^2 + 1$ and $g(x) = 2x - 3$, what is the value of $f(g(2))$?

(A) 5

(B) 1

(C) 2

(D) 3

42. Medium · No calculator needed

The function C gives the Celsius equivalent of a temperature reading x, and the function P gives the predicted crop yield, in tons, at temperature $x°$C for a greenhouse. Suppose $C(30) = 30$ and $P(30) = 18$. What is the value of $P(C(30))$?

(A) 30

(B) 18

(C) 48

(D) 540

43. Medium · Desmos

For which value of x is the function $f(x) = \dfrac{2x - 1}{x^2 - 2x - 24}$ undefined?

(A) 2

(B) 5

(C) -6

(D) -4

44. Medium · Desmos

The function h is defined by $h(x) = 3x^2 - 24x + 10$. What is the x-coordinate of the vertex of the graph of $y = h(x)$ in the xy-plane?

(A) 4

(B) 8

(C) -4

(D) 3

45. Medium · Desmos

During a scheduled maintenance check, the water level change in a water reservoir is modeled by $f(x) = x^3 - 5x + 12$, where $f(x)$ is the change in level, in centimeters, and x is the number of hours after maintenance begins, for $-3 \leq x \leq 3$. What is the value of $f(-2)$?

(A) 10

(B) 14

(C) 18

(D) 6

46. Medium · Desmos

A decorative fountain shoots a stream of water whose height, in meters, is modeled by $h(t) = -5t^2 + 20t$, where t is the time in seconds after the water leaves the nozzle. What is the maximum height, in meters, reached by the stream of water?

(A) 10

(B) 15

(C) 20

(D) 25

47. Medium · Desmos

If $f(x) = 2x^2 - 5x + 3$, what is the value of $f(4)$?

(A) 15

(B) 17

(C) 11

(D) 13

48. Medium · Desmos

A farmer has 80 meters of fencing to enclose a rectangular plot, with one side against an existing wall (so fencing is needed only on the other three sides). If x represents the width of the plot (perpendicular to the wall), which function $A(x)$ models the area of the plot, in square meters?

(A) $A(x) = 80x - x^2$

(B) $A(x) = 40x - 2x^2$

(C) $A(x) = 80 - 2x^2$

(D) $A(x) = 80x - 2x^2$

49. Medium · Desmos

The function f is defined by $f(x) = (-5)(3)^x + 7$. What is the y-intercept of the graph of $y = f(x)$ in the xy-plane?

(A) $(0, -12)$

(B) $(0, -5)$

(C) $(0, 2)$

(D) $(0, 7)$

50. Medium · Desmos

x and y^2 are directly proportional. If $x = 6$ when $y^2 = 8$, what is the value of y^2 when $x = 15$?

51. Medium · Desmos

Which of the following parabolas has its turning point in the second quadrant of the coordinate plane?

(A) $y = -(x+3)^2 - 1$

(B) $y = -(x+1)^2 + 2$

(C) $y = (x+2)^2 - 3$

(D) $y = (x-3)^2 - 1$

52. Medium · Desmos

The graph of $y = 2x^2 + 12x + 11$ in the xy-plane is a parabola. What are the coordinates of its vertex?

(A) $(3,\ 65)$

(B) $(-3,\ 11)$

(C) $(-6,\ 11)$

(D) $(-3,\ -7)$

53. Medium · Desmos

The functions f and g are defined by $f(x) = 2x - 3$ and $g(x) = x^2 + 1$. Which expression is equivalent to $g(f(x))$?

(A) $4x^2 - 12x + 10$

(B) $4x^2 - 12x + 9$

(C) $2x^2 - 1$

(D) $4x^2 + 10$

54. Medium · Desmos

The function g is defined by $g(x) = \dfrac{5}{(x-2)^2 - 9}$. For which values of x is $g(x)$ undefined?

(A) $x = 2$ only

(B) $x = -1$ and $x = 5$

(C) $x = -5$ and $x = 1$

(D) $x = 11$ only

55. Medium · Desmos

The function $h(x) = a^x + b$, where $a > 0$ and b is a constant, passes through $(0, 10)$ and $(2, 13)$. What is the value of ab?

(A) 26

(B) 13

(C) 18

(D) 20

56. Medium · Desmos

The function $h(x) = 2(x-4)^2 - 32$ has x-intercepts at $(0, 0)$ and $(t, 0)$. What is the value of t?

57. Medium · Desmos

The function g is defined by $g(x) = \dfrac{2x - 4}{(x+11)(x-6)}$. If $g(a+5) = 0$, what is the value of a?

58. HARD · DESMOS

A town's population is modeled by $P(t) = 50(3)^{\frac{t}{4}}$, where t is the number of years since 2010. According to the model, in how many years after 2010 will the population be 1,350?